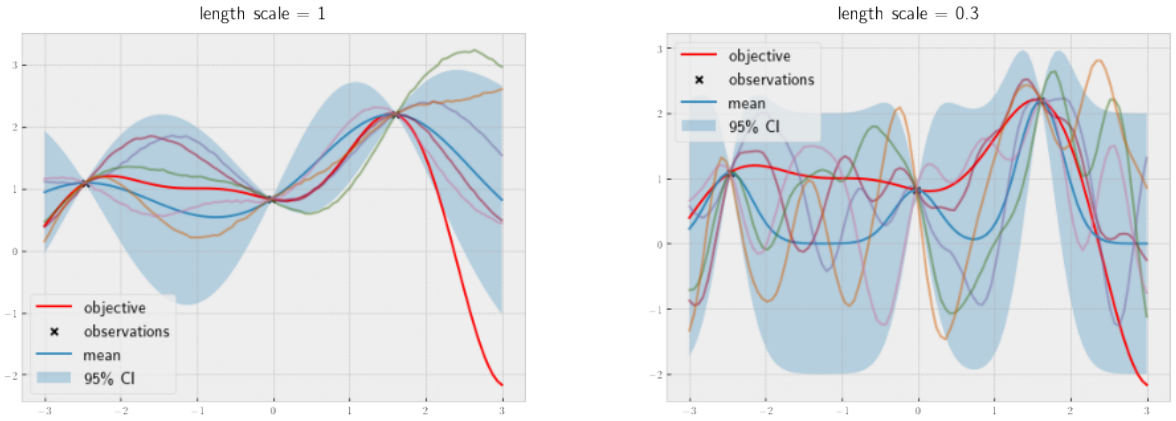


Figure 3.11 GP's predictions with the length scale set to 1 (left) and to 0.3 (right). With a small length scale, the GP predictions have more variability, leading to more uncertainty.

The stark difference between the two panels makes clear the effect of our length scale:

- A shorter length scale corresponds more variability in the objective function, given a constant change in the input.
- A longer length scale, on the other hand, forces the function to be smoother in the sense that it varies less, given the same input change.

For instance, going one unit along the x-axis, the samples in the left panel in Figure 3.11 vary less than those in the right panel.

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